

Machine Learning

CMPSCI 589

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Multinomial Naive Bayes



Review:

Document Classification

Bernoulli Naive Bayes

Review: Document Classification - Bernoulli Naive Bayes

Consider the following document, **b**, represented with the Bernoulli model:

b

	password	expired	send	review	conference	account	paper
doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those $N=7$ words in a document?

$$= \Pr(w_1 = \text{password})$$

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doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those $N=7$ words in a document?

$$= \Pr(w_1 = \text{password}) \times \Pr(w_2 = \text{expired})$$

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	password	expired	send	review	conference	account	paper
doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those $N=7$ words in a document?

$$= \Pr(w_1 = \text{password}) \times \Pr(w_2 = \text{expired}) \times (1 - \Pr(w_3 = \text{send}))$$

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What is the probability of observing exactly those $N=7$ words in a document?

$$= \Pr(w_1 = \text{password}) \times \Pr(w_2 = \text{expired}) \times (1 - \Pr(w_3 = \text{send})) \times \dots$$

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	password	expired	send	review	conference	account	paper
doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those $N=7$ words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^7 b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

The i^{th} element of document **b**
= 1 if the i^{th} word in the vocabulary appears in **b**
= 0 otherwise

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$$\Pr(\mathbf{b}) = \prod_{i=1}^7 b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

Probability that the word w_i appears on a document

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doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those **N=7** words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^7 \underbrace{b_i \times \Pr(w_i)}_{\text{if } w_i \text{ appears in the document:}} + \underbrace{(1 - b_i) \times (1 - \Pr(w_i))}_{\text{if } w_i \text{ does not appear in the document:}}$$

If w_i appears in the document:

$$\begin{aligned} &= 1 \times \Pr(w_i) \\ &= \Pr(w_i) \end{aligned} \quad \begin{aligned} &(1 - 1) \times (1 - \Pr(w_i)) \\ &= 0 \times (1 - \Pr(w_i)) \\ &= 0 \end{aligned}$$

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What is the probability of observing exactly those **N=7** words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^7 \underbrace{b_i \times \Pr(w_i)}_{= \Pr(w_i)} + \underbrace{(1 - b_i) \times (1 - \Pr(w_i))}$$

If w_i appears in
the document:

$$= \Pr(w_i)$$

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What is the probability of observing exactly those **N=7** words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^7 \underbrace{b_i \times \Pr(w_i)}_{\text{if } w_i \text{ appears}} + \underbrace{(1 - b_i) \times (1 - \Pr(w_i))}_{\text{if } w_i \text{ does not appear}}$$

If w_i does not appear
in the document:

$$\begin{aligned} &= 0 \times \Pr(w_i) & (1 - 0) \times (1 - \Pr(w_i)) \\ &= 0 & = 1 \times (1 - \Pr(w_i)) \\ & & = (1 - \Pr(w_i)) \end{aligned}$$

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doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those **N=7** words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^7 \underbrace{b_i \times \Pr(w_i)}_{\text{word appears}} + \underbrace{(1 - b_i) \times (1 - \Pr(w_i))}_{\text{word does not appear}}$$

If w_i does not appear
in the document:

$$= (1 - \Pr(w_i))$$

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doc1	1	1	0	1	0	1	0

What is the probability of observing exactly those **N=7** words in a document?

$$\Pr(\mathbf{b}) = \prod_{i=1}^7 b_i \times \Pr(w_i) + (1 - b_i) \times (1 - \Pr(w_i))$$

What is the probability of the document belonging to class y_i ?

$$\Pr(y_i | \mathbf{Doc}) = \Pr(y_i | \mathbf{b}) = \frac{\Pr(y_i) \Pr(\mathbf{b} | y_i)}{\Pr(\mathbf{b})}$$

Bayes Theorem

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What is the probability of the document belonging to class y_i ?

$$\Pr(y_i | \mathbf{Doc}) \propto \Pr(y_i | \mathbf{b}) = \Pr(y_i) \Pr(\mathbf{b} | y_i)$$

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What is the probability of the document belonging to class y_i ?

$$\begin{aligned}\Pr(y_i | \mathbf{Doc}) &\propto \Pr(y_i | \mathbf{b}) = \underbrace{\Pr(y_i) \Pr(\mathbf{b} | y_i)} \\ &= \underbrace{\Pr(y_i)}\end{aligned}$$

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What is the probability of the document belonging to class y_i ?

$$\begin{aligned}\Pr(y_i | \mathbf{Doc}) &\propto \Pr(y_i | \mathbf{b}) = \Pr(y_i) \underbrace{\Pr(\mathbf{b} | y_i)} \\ &= \Pr(y_i) \prod_{k=1}^7 b_k \times \Pr(w_k | y_i) + (1 - b_k) \times (1 - \Pr(w_k | y_i))\end{aligned}$$

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What is the probability of the document belongs to class y_i ?

$$\Pr(y_i | \mathbf{b}) = \Pr(y_i) \prod_{k=1}^7 b_k \times \Pr(w_k | y_i) + (1 - b_k) \times (1 - \Pr(w_k | y_i))$$

More generally:

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

- $|V|$ is the size of the vocabulary (i.e., the size of **b**)
- b_k is an entry in **b** representing whether the k^{th} word shows up in the document (0 or 1)
- $\Pr(w_k | y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i
- $1 - \Pr(w_k | y_i)$ is the probability of the k^{th} word **not** appearing in documents of the class y_i

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Consider the following document, **b**, represented with the Bernoulli model:

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What is the probability of the document belongs to class y_i ?

$$\Pr(y_i | \mathbf{b}) = \Pr(y_i) \prod_{k=1}^7 b_k \times \Pr(w_k | y_i) + (1 - b_k) \times (1 - \Pr(w_k | y_i))$$

More generally:

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} [b_k \times \Pr(w_k | y_i) + (1 - b_k) \times (1 - \Pr(w_k | y_i))]$$

How to estimate these probabilities based on training data?

- $\Pr(w_k | y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i
- $1 - \Pr(w_k | y_i)$ is the probability of the k^{th} word **not** appearing in documents of the class y_i

Document Classification - Bernoulli Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

- V is the vocabulary of all known words
- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i
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Document Classification - Bernoulli Naive Bayes

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$$\Pr(y_i) = \frac{N(y_i)}{N}$$

percentage of
documents
of class y_i

Document Classification - Bernoulli Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

- V is the vocabulary of all known words
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$$\Pr(y_i) = \frac{N(y_i)}{N}$$

percentage of
documents
of class y_i

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

percentage of
documents of class y_i
that contain the word w_k

Bernoulli Naive Bayes Classification

- Consider the problem of classifying a document
 - it may belong to the “Sports” class (**SP**), or to the “Informatics” (**INF**) class...
 - depending on the topics being discussed in the document
- We analyzed the training set (composed of many documents) and identified the following vocabulary

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix} \quad |V| = 8$$

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) $\rightarrow V =$

$$\left[\begin{array}{l} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{array} \right]$$

$$\mathbf{B}_{\text{SP}} = \begin{matrix} & \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{matrix} \\ \left(\begin{array}{cccccccc} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{array} \right. \end{matrix}$$

Number of documents belonging to the class **SP** ("Sports")

Number of words in the vocabulary
(considering documents of all classes)

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) $\rightarrow$

$$v = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\mathbf{B}_{\text{SP}} = \begin{matrix} & \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{matrix} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} & \left. \vphantom{\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}} \right\} \text{6 documents} \end{matrix}$$

$$\mathbf{B}_{\text{INF}} = \begin{matrix} & \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{matrix} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} & \left. \vphantom{\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}} \right\} \text{5 documents} \end{matrix}$$

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) $\rightarrow$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\mathbf{B}_{\text{SP}} = \begin{matrix} & \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{matrix} \\ \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix} & \left. \vphantom{\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}} \right\} \begin{matrix} \text{How often the word } w_3 \\ \text{appears in all documents} \\ \\ \text{6 documents} \end{matrix} \end{matrix}$$

$$\mathbf{B}_{\text{INF}} = \begin{matrix} & \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{matrix} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} & \left. \vphantom{\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}} \right\} \begin{matrix} \\ \\ \\ \\ \text{5 documents} \end{matrix} \end{matrix}$$

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) $\rightarrow$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

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Description of the 3rd document belonging to the class "Sports"

6 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

5 documents

Bernoulli Naive Bayes Classification

Attributes of a document (words that may appear in it) $\rightarrow$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

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$$\mathbf{B}_{\text{INF}} = \begin{matrix} & \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} & \left. \vphantom{\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}} \right\} \text{5 documents} \end{matrix}$$

Description of the 2nd document belonging to the class "Informatics"

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \underbrace{\Pr(y_i)} \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k) (1 - \Pr(w_k \mid y_i)) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(y_i) = \frac{N(y_i)}{N} \quad \begin{array}{l} \text{percentage of} \\ \text{documents} \\ \text{of class } y_i \end{array}$$

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[\underbrace{b_k}_{\text{prior}} \underbrace{\Pr(w_k \mid y_i)}_{\text{likelihood}} + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$
w_1	
w_2	
w_3	
w_4	
w_5	
w_6	
w_7	
w_8	

	$n(\cdot, \text{INF})$

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[\underbrace{b_k}_{\text{smoothed prior}} \underbrace{\Pr(w_k | y_i)}_{\text{word probability}} + (1 - b_k)(1 - \Pr(w_k | y_i)) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k | y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$
w_1	3
w_2	
w_3	
w_4	
w_5	
w_6	
w_7	
w_8	

$n(\cdot, \text{INF})$

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[\underbrace{b_k \Pr(w_k \mid y_i)} + \underbrace{(1 - b_k)(1 - \Pr(w_k \mid y_i))} \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k \mid y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$
w_1	3
w_2	1
w_3	
w_4	
w_5	
w_6	
w_7	
w_8	

$n(\cdot, \text{INF})$

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[\underbrace{b_k}_{\text{prior}} \underbrace{\Pr(w_k | y_i)}_{\text{likelihood}} + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k | y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$
w_1	3
w_2	1
w_3	2
w_4	
w_5	
w_6	
w_7	
w_8	

$n(\cdot, \text{INF})$

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[\underbrace{b_k}_{\text{prior}} \underbrace{\Pr(w_k | y_i)}_{\text{likelihood}} + (1 - b_k)(1 - \Pr(w_k | y_i)) \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k | y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$
w_1	3
w_2	1
w_3	2
w_4	3
w_5	3
w_6	4
w_7	4
w_8	4

$n(\cdot, \text{INF})$
1

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[\underbrace{b_k \Pr(w_k | y_i)} + \underbrace{(1 - b_k)(1 - \Pr(w_k | y_i))} \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k | y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$
w_1	3
w_2	1
w_3	2
w_4	3
w_5	3
w_6	4
w_7	4
w_8	4

$n(\cdot, \text{INF})$
1
3

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[\underbrace{b_k \Pr(w_k | y_i)} + (1 - b_k) \underbrace{(1 - \Pr(w_k | y_i))} \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k | y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$
w_1	3
w_2	1
w_3	2
w_4	3
w_5	3
w_6	4
w_7	4
w_8	4

$n(\cdot, \text{INF})$
1
3
3

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[\underbrace{b_k \Pr(w_k | y_i)} + (1 - b_k) \underbrace{(1 - \Pr(w_k | y_i))} \right]$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{N(y_i)}$$

- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the number of documents belonging to class y_i where the word w_k appears
- $\Pr(w_k | y_i)$ is the probability of the k^{th} word appearing in documents of the class y_i

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

6 documents

$$\mathbf{B}_{\text{SP}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{B}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 6/11 \times$$

$$(1-0) (1 - 3/6) \times$$

$$1 \quad 1/6 \quad \times$$

$$1 \quad 2/6 \quad \times$$

$$(1-0) (1 - 3/6) \times$$

$$1 \quad 3/6 \quad \times$$

$$(1-0) (1 - 4/6) \times$$

$$1 \quad 4/6 \quad \times$$

$$(1-0) (1 - 4/6)$$

$$= 0.00028$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} | \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 1/6 \times \\ &\quad 1 \quad 2/6 \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 3/6 \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \quad 4/6 \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} \mid \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \times \frac{1}{6} \times \\ &\quad 1 \times \frac{2}{6} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \times \frac{3}{6} \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \times \frac{4}{6} \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} \mid \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 1/6 \times \\ &\quad 1 \quad 2/6 \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 3/6 \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \quad 4/6 \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} \mid \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 1/6 \times \\ &\quad 1 \quad 2/6 \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 3/6 \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \quad 4/6 \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} | \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 1/6 \times \\ &\quad 1 \quad 2/6 \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 3/6 \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \quad 4/6 \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} \mid \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 1/6 \times \\ &\quad 1 \quad 2/6 \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 3/6 \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \quad 4/6 \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} | \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 1/6 \times \\ &\quad 1 \quad 2/6 \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 3/6 \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \quad 4/6 \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$ To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} \mid \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 1/6 \times \\ &\quad 1 \quad 2/6 \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 3/6 \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \quad 4/6 \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\begin{aligned} \Pr(y = \text{SP} \mid \mathbf{b}) &= \frac{6}{11} \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 1/6 \times \\ &\quad 1 \quad 2/6 \times \\ &\quad (1-0) (1 - 3/6) \times \\ &\quad 1 \quad 3/6 \times \\ &\quad (1-0) (1 - 4/6) \times \\ &\quad 1 \quad 4/6 \times \\ &\quad (1-0) (1 - 4/6) \\ &= 0.00028 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} \mid \mathbf{b}) = 0.00028$$

$$\Pr(y = \text{INF} \mid \mathbf{b}) = \frac{5}{11} \times (1-0) (1 - 1/5) \times 1 \times \frac{3}{5} \times 1 \times \frac{3}{5} \times (1-0) (1 - 1/5) \times 1 \times \frac{1}{5} \times (1-0) (1 - 1/5) \times 1 \times \frac{3}{5} \times (1-0) (1 - 1/5) = 0.008$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} | \mathbf{b}) &= \frac{5}{11} \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 1/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \\ &= 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k)(1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} | \mathbf{b}) &= \frac{5}{11} \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \times \frac{3}{5} \times \\ &\quad 1 \times \frac{3}{5} \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \times \frac{1}{5} \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \times \frac{3}{5} \times \\ &\quad (1-0) (1 - 1/5) \\ &= 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} | \mathbf{b}) = & \frac{5}{11} \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 3/5 \times \\ & 1 \quad 3/5 \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 1/5 \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 3/5 \times \\ & (1-0) (1 - 1/5) \\ & = 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} \mid \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} \mid \mathbf{b}) = & \frac{5}{11} \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 3/5 \times \\ & 1 \quad 3/5 \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 1/5 \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 3/5 \times \\ & (1-0) (1 - 1/5) \\ & = 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k)(1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} | \mathbf{b}) = & \frac{5}{11} \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 3/5 \times \\ & 1 \quad 3/5 \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 1/5 \times \\ & (1-0) (1 - 1/5) \times \\ & 1 \quad 3/5 \times \\ & (1-0) (1 - 1/5) \\ & = 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} | \mathbf{b}) &= \frac{5}{11} \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 1/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \\ &= 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} | \mathbf{b}) &= \frac{5}{11} \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 1/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \\ &= 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k)(1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$ To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} | \mathbf{b}) &= \frac{5}{11} \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 1/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \\ &= 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k | y_i) + (1 - b_k) (1 - \Pr(w_k | y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} | \mathbf{b}) = 0.00028$$

$$\begin{aligned} \Pr(y = \text{INF} | \mathbf{b}) &= \frac{5}{11} \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 1/5 \times \\ &\quad (1-0) (1 - 1/5) \times \\ &\quad 1 \quad 3/5 \times \\ &\quad (1-0) (1 - 1/5) \\ &\quad = 0.008 \end{aligned}$$

Bernoulli Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{|V|} \left[b_k \Pr(w_k \mid y_i) + (1 - b_k)(1 - \Pr(w_k \mid y_i)) \right]$$

Consider a new document described by $\mathbf{b} = (0, 1, 1, 0, 1, 0, 1, 0)$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	3	3/6	1	1/5
w_2	1	1/6	3	3/5
w_3	2	2/6	3	3/5
w_4	3	3/6	1	1/5
w_5	3	3/6	1	1/5
w_6	4	4/6	1	1/5
w_7	4	4/6	3	3/5
w_8	4	4/6	1	1/5

$$\Pr(y = \text{SP} \mid \mathbf{b}) = 0.00028$$

$$\Pr(y = \text{INF} \mid \mathbf{b}) = 0.008$$



The new document seems to belong to the “Informatics” class

Document Classification: Multinomial Naive Bayes

Multinomial Naive Bayes Classification

- **Multinomial model:** a document is represented by a vector with integer components
each entry indicates the frequency of a word in the document

$\mathbf{m} = (2, 1, 0, 1, 0, 1, 0)$

	password	expired	send	review	conference	account	paper
doc1	2	1	0	1	0	1	0

- To classify a new document, compute the following for each class y_i :

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

where

- $\text{len}(\text{Doc})$ is the length of document Doc
→ defined as the number of (unique) words that appear in that document
- $\Pr(w_k \mid y_i)$ is the relative frequency of the k^{th} word in documents of the class y_i

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- $\text{len}(\text{Doc})$ is the length of Doc (i.e., the number of unique words that appear in that document)
- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the frequency of the word w_k in documents belonging to class y_i

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- $\text{len}(\text{Doc})$ is the length of Doc (i.e., the number of unique words that appear in that document)
- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the frequency of the word w_k in documents belonging to class y_i

$$\Pr(y_i) = \frac{N(y_i)}{N}$$

percentage of
documents
of class y_i

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- $\text{len}(\text{Doc})$ is the length of Doc (i.e., the number of unique words that appear in that document)
- N is the total number of documents available
- $N(y_i)$ is the number of documents belonging to class y_i
- $n(w_k, y_i)$ is the frequency of the word w_k in documents belonging to class y_i

$$\Pr(y_i) = \frac{N(y_i)}{N}$$

percentage of
documents
of class y_i

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

total number of times word w_k
appears in documents of class y_i

divided by

total number of words that
appear in documents of class y_i



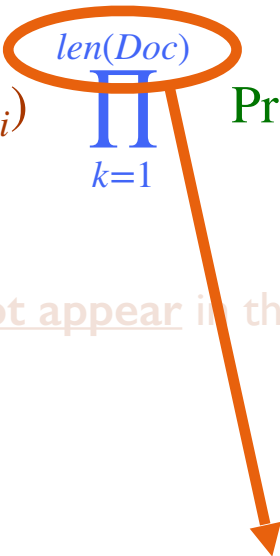
the relative frequency of word w_k in documents of class y_i

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- **Multinomial** NB ignores words that do not appear in the document being classified

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$


- **Multinomial** NB ignores words that do not appear in the document being classified

Length of document *Doc*

Defined as the number of (*unique*) words that **do** appear in that document

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- **Multinomial** NB ignores words that do not appear in the document being classified
 - (unlike Bernoulli, which considers all words in the vocabulary, whether they appear or not)

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- **Multinomial** NB ignores words that do not appear in the document being classified
 - (unlike Bernoulli, which considers all words in the vocabulary, whether they appear or not)

In practice, only words that appear in the document being classified need to be taken into account

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- **Multinomial** NB ignores words that do not appear in the document being classified
 - (unlike Bernoulli, which considers all words in the vocabulary, whether they appear or not)
- **Binomial** NB cares about the absence and presence of words in the document
 - It penalizes a class when a word common to that class is absent from the document
 - Useful if the absence of words is informative of the class of the document

Document Classification - Multinomial Naive Bayes

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

- **Multinomial** NB ignores words that do not appear in the document being classified
 - (unlike Bernoulli, which considers all words in the vocabulary, whether they appear or not)
- **Binomial** NB cares about the absence and presence of words in the document
 - It penalizes a class when a word common to that class is absent from the document
 - Useful if the absence of words is informative of the class of the document
- **Multinomial** NB has no such penalty for absent words
 - Generally better for longer documents ...
 - where word frequency is more informative of the document's likely class

Multinomial Naive Bayes Classification

- Let us revisit the problem of classifying a document
 - It may belong to the “Sports” class (**SP**), or to the “Informatics” (**INF**) class...
 - Depending on the topics being discussed in the document

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix} \quad |V| = 8$$

Multinomial Naive Bayes Classification

Attributes of a document (words that may appear in it) $\rightarrow$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\mathbf{M}_{\text{SP}} = \begin{matrix} & \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{matrix} \\ \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix} & \left. \vphantom{\begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}} \right\} \text{6 documents} \end{matrix}$$

$$\mathbf{M}_{\text{INF}} = \begin{matrix} & \begin{matrix} w_1 & w_2 & w_3 & w_4 & w_5 & w_6 & w_7 & w_8 \end{matrix} \\ \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} & \left. \vphantom{\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}} \right\} \text{5 documents} \end{matrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \underbrace{\Pr(y_i)} \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(y_i) = \frac{N(y_i)}{N} \quad \begin{array}{l} \text{percentage of} \\ \text{documents} \\ \text{of class } y_i \end{array}$$

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k | y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

→ total number of times word w_k appears in documents of class y_i
 → total number of words that appear in documents of class y_i

	$n(\cdot, \text{SP})$
w_1	
w_2	
w_3	
w_4	
w_5	
w_6	
w_7	
w_8	

	$n(\cdot, \text{INF})$

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k | y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

→ total number of times word w_k appears in documents of class y_i
 → total number of words that appear in documents of class y_i

	$n(\cdot, \text{SP})$
w_1	5
w_2	
w_3	
w_4	
w_5	
w_6	
w_7	
w_8	

$n(\cdot, \text{INF})$

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

→ total number of times word w_k appears in documents of class y_i
 → total number of words that appear in documents of class y_i

	$n(\cdot, \text{SP})$
w_1	5
w_2	1
w_3	
w_4	
w_5	
w_6	
w_7	
w_8	

$n(\cdot, \text{INF})$

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k | y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

→ total number of times word w_k appears in documents of class y_i
 → total number of words that appear in documents of class y_i

	$n(\cdot, \text{SP})$
w_1	5
w_2	1
w_3	2
w_4	
w_5	
w_6	
w_7	
w_8	

$n(\cdot, \text{INF})$

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k | y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

→ total number of times word w_k appears in documents of class y_i
 → total number of words that appear in documents of class y_i

	$n(\cdot, \text{SP})$
w_1	5
w_2	1
w_3	2
w_4	5
w_5	4
w_6	6
w_7	7
w_8	6

$n(\cdot, \text{INF})$
1

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i | \text{Doc})$

$$\Pr(y_i | \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k | y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

→ total number of times word w_k appears in documents of class y_i
 → total number of words that appear in documents of class y_i

	$n(\cdot, \text{SP})$
w_1	5
w_2	1
w_3	2
w_4	5
w_5	4
w_6	6
w_7	7
w_8	6

$n(\cdot, \text{INF})$
1
4

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

→ total number of times word w_k appears in documents of class y_i
 → total number of words that appear in documents of class y_i

	$n(\cdot, \text{SP})$
w_1	5
w_2	1
w_3	2
w_4	5
w_5	4
w_6	6
w_7	7
w_8	6

$n(\cdot, \text{INF})$
1
4
3
1
1
2
3
1

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

Estimating the probabilities needed to compute $\Pr(y_i \mid \text{Doc})$

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

$$V = \begin{bmatrix} w_1 = \text{goal}, \\ w_2 = \text{tutor}, \\ w_3 = \text{variance}, \\ w_4 = \text{speed}, \\ w_5 = \text{drink}, \\ w_6 = \text{defence}, \\ w_7 = \text{performance}, \\ w_8 = \text{field} \end{bmatrix}$$

$$\Pr(w_k \mid y_i) = \frac{n(w_k, y_i)}{\sum_{s=1}^{|V|} n(w_s, y_i)}$$

→ total number of times word w_k appears in documents of class y_i

→ total number of words that appear in documents of class y_i

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

6 documents

$$\mathbf{M}_{\text{SP}} = \begin{pmatrix} 2 & 0 & 0 & 0 & 1 & 2 & 3 & 1 \\ 0 & 0 & 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 2 & 1 & 0 \\ 1 & 0 & 0 & 2 & 0 & 1 & 0 & 1 \\ 2 & 0 & 0 & 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}$$

5 documents

$$\mathbf{M}_{\text{INF}} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 6/11 \times$$

$$\Pr(w_3 \mid \text{SP}) \times$$

$$\Pr(w_5 \mid \text{SP}) \times$$

$$\Pr(w_2 \mid \text{SP}) \times$$

$$\Pr(w_7 \mid \text{SP})$$

$$= 6/11 \times$$

$$2/36 \times$$

$$4/36 \times$$

$$1/36 \times$$

$$7/36$$

$$= 0.0000181$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = \frac{6}{11} \times$$

$$\Pr(w_3 \mid \text{SP}) \times$$

$$\Pr(w_5 \mid \text{SP}) \times$$

$$\Pr(w_2 \mid \text{SP}) \times$$

$$\Pr(w_7 \mid \text{SP})$$

$$= \frac{6}{11} \times$$

$$\frac{2}{36} \times$$

$$\frac{4}{36} \times$$

$$\frac{1}{36} \times$$

$$\frac{7}{36}$$

$$= 0.0000181$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

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$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
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w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 6/11 \times$$

$$\Pr(w_3 \mid \text{SP}) \times$$

$$\Pr(w_5 \mid \text{SP}) \times$$

$$\Pr(w_2 \mid \text{SP}) \times$$

$$\Pr(w_7 \mid \text{SP})$$

$$= 6/11 \times$$

$$2/36 \times$$

$$4/36 \times$$

$$1/36 \times$$

$$7/36$$

$$= 0.0000181$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 6/11 \times$$

$$\Pr(w_3 \mid \text{SP}) \times$$

$$\Pr(w_5 \mid \text{SP}) \times$$

$$\Pr(w_2 \mid \text{SP}) \times$$

$$\Pr(w_7 \mid \text{SP})$$

$$= 6/11 \times$$

$$2/36 \times$$

$$4/36 \times$$

$$1/36 \times$$

$$7/36$$

$$= 0.0000181$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 6/11 \times$$

$$\Pr(w_3 \mid \text{SP}) \times$$

$$\Pr(w_5 \mid \text{SP}) \times$$

$$\Pr(w_2 \mid \text{SP}) \times$$

$$\Pr(w_7 \mid \text{SP})$$

$$= 6/11 \times$$

$$2/36 \times$$

$$4/36 \times$$

$$1/36 \times$$

$$7/36$$

$$= 0.0000181$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\begin{aligned}
 \Pr(y = \text{SP} \mid \text{Doc}) &= 6/11 \times \\
 &\quad \Pr(w_3 \mid \text{SP}) \times \\
 &\quad \Pr(w_5 \mid \text{SP}) \times \\
 &\quad \Pr(w_2 \mid \text{SP}) \times \\
 &\quad \Pr(w_7 \mid \text{SP}) \\
 &= 6/11 \times \\
 &\quad 2/36 \times \\
 &\quad 4/36 \times \\
 &\quad 1/36 \times \\
 &\quad 7/36 \\
 &= 0.0000181
 \end{aligned}$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\begin{aligned} \Pr(y = \text{INF} \mid \text{Doc}) &= 5/11 \times \\ &\quad \Pr(w_3 \mid \text{INF}) \times \\ &\quad \Pr(w_5 \mid \text{INF}) \times \\ &\quad \Pr(w_2 \mid \text{INF}) \times \\ &\quad \Pr(w_7 \mid \text{INF}) \\ &= 5/11 \times \\ &\quad 3/16 \times \\ &\quad 1/16 \times \\ &\quad 4/16 \times \\ &\quad 3/16 \\ &= 0.000249 \end{aligned}$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\Pr(y = \text{INF} \mid \text{Doc}) = \frac{5}{11} \times$$

$$\Pr(w_3 \mid \text{INF}) \times$$

$$\Pr(w_5 \mid \text{INF}) \times$$

$$\Pr(w_2 \mid \text{INF}) \times$$

$$\Pr(w_7 \mid \text{INF})$$

$$= \frac{5}{11} \times$$

$$\frac{3}{16} \times$$

$$\frac{1}{16} \times$$

$$\frac{4}{16} \times$$

$$\frac{3}{16}$$

$$= 0.000249$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\Pr(y = \text{INF} \mid \text{Doc}) = \frac{5}{11} \times$$

$$\Pr(w_3 \mid \text{INF}) \times$$

$$\Pr(w_5 \mid \text{INF}) \times$$

$$\Pr(w_2 \mid \text{INF}) \times$$

$$\Pr(w_7 \mid \text{INF})$$

$$= \frac{5}{11} \times$$

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$$\frac{1}{16} \times$$

$$\frac{4}{16} \times$$

$$\frac{3}{16}$$

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Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\begin{aligned} \Pr(y = \text{INF} \mid \text{Doc}) &= \frac{5}{11} \times \\ &\quad \Pr(w_3 \mid \text{INF}) \times \\ &\quad \Pr(w_5 \mid \text{INF}) \times \\ &\quad \Pr(w_2 \mid \text{INF}) \times \\ &\quad \Pr(w_7 \mid \text{INF}) \\ &= \frac{5}{11} \times \\ &\quad \frac{3}{16} \times \\ &\quad \frac{1}{16} \times \\ &\quad \frac{4}{16} \times \\ &\quad \frac{3}{16} \\ &= 0.000249 \end{aligned}$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\begin{aligned} \Pr(y = \text{INF} \mid \text{Doc}) &= \frac{5}{11} \times \\ &\quad \Pr(w_3 \mid \text{INF}) \times \\ &\quad \Pr(w_5 \mid \text{INF}) \times \\ &\quad \Pr(w_2 \mid \text{INF}) \times \\ &\quad \Pr(w_7 \mid \text{INF}) \\ &= \frac{5}{11} \times \\ &\quad \frac{3}{16} \times \\ &\quad \frac{1}{16} \times \\ &\quad \frac{4}{16} \times \\ &\quad \frac{3}{16} \\ &= 0.000249 \end{aligned}$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
w_4	5	5/36	1	1/16
w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\begin{aligned} \Pr(y = \text{INF} \mid \text{Doc}) &= \frac{5}{11} \times \\ &\quad \Pr(w_3 \mid \text{INF}) \times \\ &\quad \Pr(w_5 \mid \text{INF}) \times \\ &\quad \Pr(w_2 \mid \text{INF}) \times \\ &\quad \Pr(w_7 \mid \text{INF}) \\ &= \frac{5}{11} \times \\ &\quad \frac{3}{16} \times \\ &\quad \frac{1}{16} \times \\ &\quad \frac{4}{16} \times \\ &\quad \frac{3}{16} \\ &= 0.000249 \end{aligned}$$

Multinomial Naive Bayes Classification

$$\Pr(y_i \mid \text{Doc}) = \Pr(y_i) \prod_{k=1}^{\text{len}(\text{Doc})} \Pr(w_k \mid y_i)$$

Consider a new document containing the words $\{w_3, w_5, w_2, w_7\}$. To which class does it belong?

$$\Pr(y_i = \text{SP}) = 6/11$$

$$\Pr(y_i = \text{INF}) = 5/11$$

	$n(\cdot, \text{SP})$	$\Pr(\cdot \mid y = \text{SP})$	$n(\cdot, \text{INF})$	$\Pr(\cdot \mid y = \text{INF})$
w_1	5	5/36	1	1/16
w_2	1	1/36	4	4/16
w_3	2	2/36	3	3/16
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w_5	4	4/36	1	1/16
w_6	6	6/36	2	2/16
w_7	7	7/36	3	3/16
w_8	6	6/36	1	1/16

$$\Pr(y = \text{SP} \mid \text{Doc}) = 0.0000181$$

$$\Pr(y = \text{INF} \mid \text{Doc}) = 0.000249$$



The new document seems to belong to the "Informatics" class

Laplace Smoothing

- Bernoulli/Multinomial Naive Bayes are affected by attributes whose values never show up in the dataset
 - Similar to the problem discussed in the previous lecture
 - “we’re trying to predict something about a red car, but no red cars appeared in the dataset”
 - This is the zero frequency problem
- Here, the problem is caused by a word that appears in the document being classified
 - But that was never observed in the dataset, in documents of a given class y_i
 - In that case, the probability of that class will be zero

Solution is similar to the one presented earlier:

(intuitively, adjust the numerator and denominator so the division is not equal to zero)

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i) + 1}{N(y_i) + |V|}$$

Bernoulli Model

$$\Pr(w_k | y_i) = \frac{n(w_k, y_i) + 1}{\sum_{s=1}^{|V|} n(w_s, y_i) + |V|}$$

Multinomial Model

Bernoulli vs Multinomial Naive Bayes

- **Multinomial Naive Bayes** takes into account the **frequency of words**
 - but does not model how they affect each other (independence assumption)
- **Bernoulli Naive Bayes** often works better when analyzing **short documents**
- **Multinomial Naive Bayes** often works better when analyzing **long documents**
 - since it only needs to estimate the probability of words that appear in the document
- **Pre-processing the text** in the documents is important
 - **Stemming**: removes the suffix of words so they are treated as being equal
 - Fishing, fisher, fished → fish
 - Removing "stop words" → common words such as "the", "of", etc.



Machine Learning

CMPSCI 589

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Multinomial Naive Bayes

